

LOAN APPROVAL PREDICTION

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Introduction

Loan approval is a major task for financial institutions. Before approving a loan, banks must check whether the applicant is capable of repaying it. Traditionally, this process is done manually, which is slow and inefficient.

Machine Learning helps automate this process by analyzing historical applicant data and predicting loan approval status.

- ▶ Banks receive many loan applications daily. Manual verification is slow, time-consuming, and may lead to mistakes. The problem is to predict whether a loan application should be approved or rejected using machine learning.

Problem Statement

- ▶ Traditional loan approval requires manual verification.
- ▶ Large number of applications increases workload.
- ▶ Human errors can affect decision making.
- ▶ Need for an intelligent prediction system.
- ▶ Proposed Solution
- ▶ Build a machine learning model that predicts loan approval based on applicant information.

Objectives

Project Objectives

- ▶ Analyze loan applicant data.
- ▶ Perform data preprocessing.
- ▶ Conduct Exploratory Data Analysis.
- ▶ Build machine learning models.
- ▶ Compare model performance.
- ▶ Optimize model using hyperparameter tuning.
- ▶ To predict loan approval status accurately.
- ▶ Deploy the system using Streamlit.

Dataset Description

1.Dataset Name: Loan
Approval Prediction

2.Datasource: Kaggle

3.Dataset Size:

Rows:614

Columns:13

Features Used:

1. Loan_id
2. Property_area
3. Loan_status
4. Gender
5. Married
6. Dependents
7. Education
8. Self_employed
9. Applicant income
10. Co applicant income
- 11.. Loan_amount_term
12. Credit_history
13. Loan amount

Dataset

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	loan_id	gender	married	dependent	education	self_empt	applicantir	coapplicantr	loanamou	loan_amou	credit_hist	property_a	loan_status
2	lp001002	male	no	0	graduate	no	5849	0		360	1	urban	y
3	lp001003	male	yes	1	graduate	no	4583	1508	128	360	1	rural	n
4	lp001005	male	yes	0	graduate	yes	3000	0	66	360	1	urban	y
5	lp001006	male	yes	0	not gradua	no	2583	2358	120	360	1	urban	y
6	lp001008	male	no	0	graduate	no	6000	0	141	360	1	urban	y
7	lp001011	male	yes	2	graduate	yes	5417	4196	267	360	1	urban	y
8	lp001013	male	yes	0	not gradua	no	2333	1516	95	360	1	urban	y
9	lp001014	male	yes	3+	graduate	no	3036	2504	158	360	0	semiurban	n
10	lp001018	male	yes	2	graduate	no	4006	1526	168	360	1	urban	y
11	lp001020	male	yes	1	graduate	no	12841	10968	349	360	1	semiurban	n
12	lp001024	male	yes	2	graduate	no	3200	700	70	360	1	urban	y
13	lp001027	male	yes	2	graduate		2500	1840	109	360	1	urban	y
14	lp001028	male	yes	2	graduate	no	3073	8106	200	360	1	urban	y
15	lp001029	male	no	0	graduate	no	1853	2840	114	360	1	rural	n
16	lp001030	male	yes	2	graduate	no	1299	1086	17	120	1	urban	y
17	lp001032	male	no	0	graduate	no	4950	0	125	360	1	urban	y
18	lp001034	male	no	1	not gradua	no	3596	0	100	240		urban	y
19	lp001036	female	no	0	graduate	no	3510	0	76	360	0	urban	n
20	lp001038	male	yes	0	not gradua	no	4887	0	133	360	1	rural	n
21	lp001041	male	yes	0	graduate		2600	3500	115		1	urban	y
22	lp001043	male	yes	0	not gradua	no	7660	0	104	360	0	urban	n
23	lp001046	male	yes	1	graduate	no	5955	5625	315	360	1	urban	y
24	lp001047	male	yes	0	not gradua	no	2600	1911	116	360	0	semiurban	n
25	lp001050		yes	2	not gradua	no	3365	1917	112	360	0	rural	n
26	lp001052	male	yes	1	graduate		3717	2925	151	360		semiurban	n
27	lp001066	male	yes	0	graduate	yes	9560	0	191	360	1	semiurban	y
28	lp001068	male	yes	0	graduate	no	2799	2253	122	360	1	semiurban	y

Technologies Used

- ▶ **Programming Language: Python**
- ▶ **Dataset Source: Kaggle**
- ▶ **Platforms Used: Google Colab, Streamlit**
- ▶ **Libraries Used: Matplotlib, Numpy, Pandas, Seaborn, Scikit-learn**
- ▶ **Machine Learning Algorithm: Decision Tree, Logistic Regression, Random Forest**
- ▶ **Visualizations: Bar Chart, Histogram, Heatmap**

Data Preprocessing

- 1.Dataset was loaded using Pandas
- 2.Checked dataset values,shape,info and missing values
- 3.Removed duplicate values
- 4.Converted Categorical data to numerical data using Label encoding
- 5.Visualized the data for more insights
- 6.Removed Unnecessary Columns

Exploratory Data Analysis

- ▶ Analyzed dataset structure and feature distributions.
- ▶ Examined relationships between variables using visualizations.
- ▶ Generated:
 - ▶ Loan Status Distribution
 - ▶ Gender Distribution
 - ▶ Education Distribution
 - ▶ Property Area Distribution
 - ▶ Applicant Income Histogram
 - ▶ Correlation Heatmap
- ▶ Key Findings
 - ▶ Loan approvals were higher than rejections.
 - ▶ Majority of applicants were graduates.
 - ▶ Semiurban property area had the highest number of applications.
 - ▶ Credit History showed the strongest impact on loan approval.

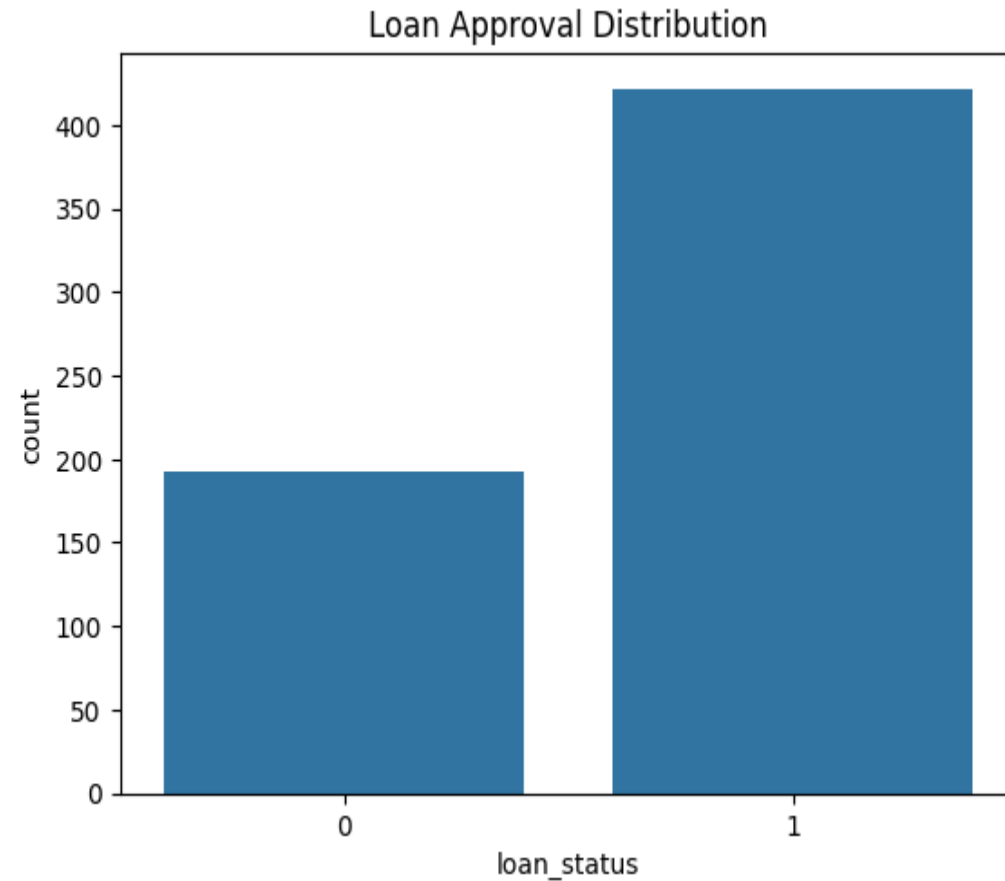
Loan Approval Distribution

Insights:

1.The number of approved loans (1) is higher than rejected loans (0).

2.This shows that most applicants in the dataset were eligible for loan approval.

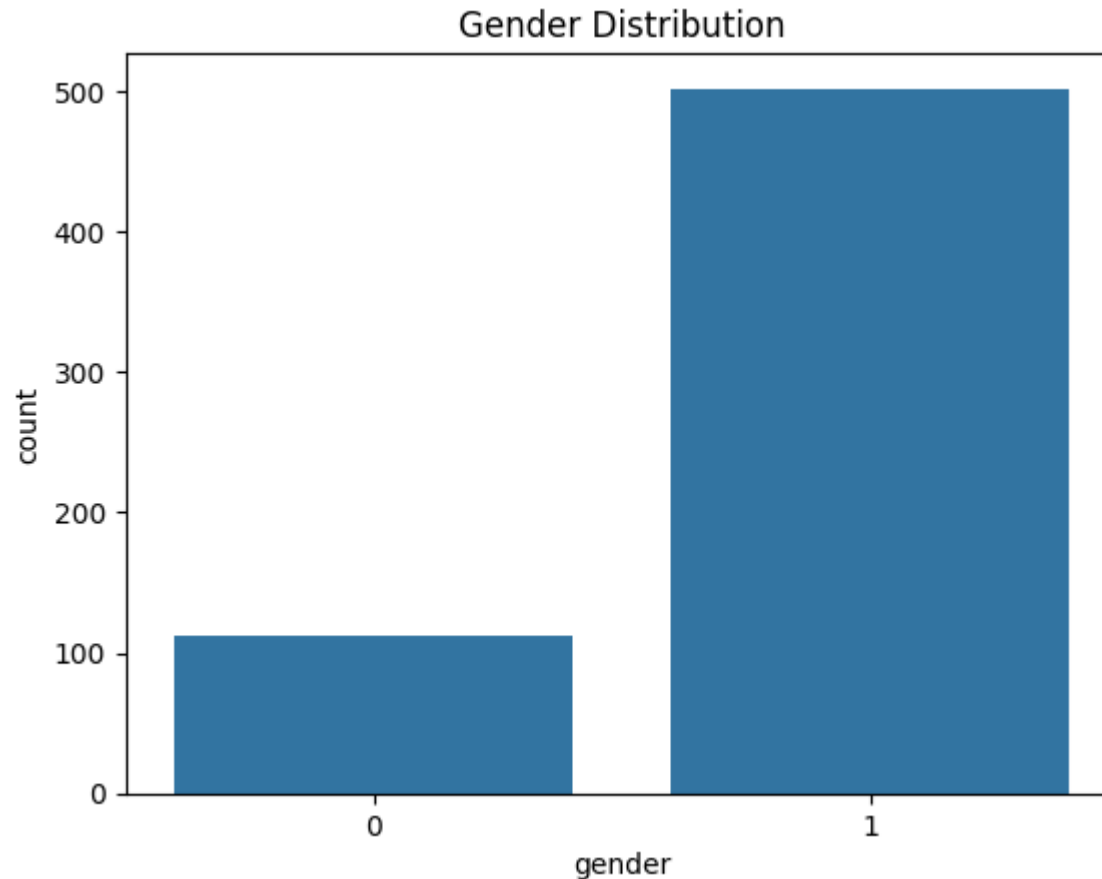
3.It also indicates that the dataset is slightly imbalanced toward approved loans.



Gender Distribution

Insights:

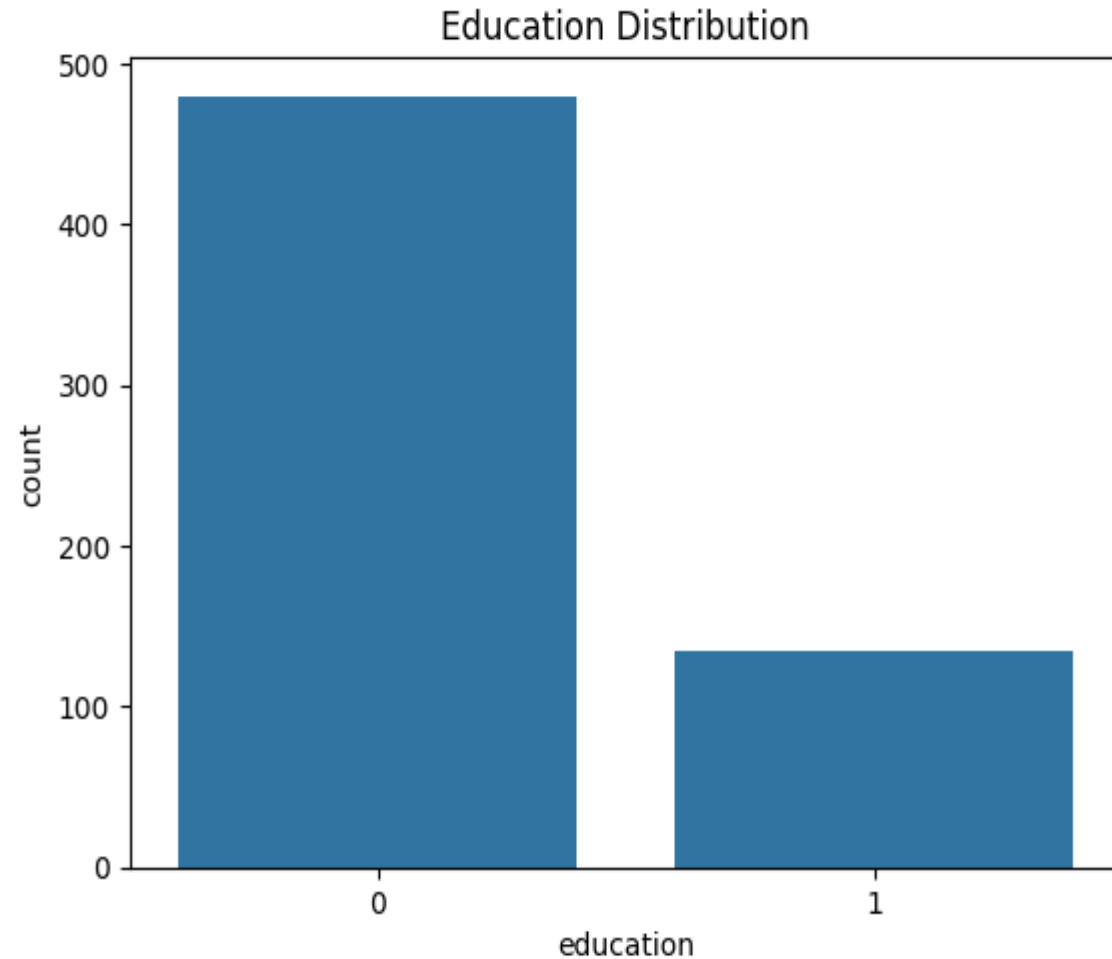
- 1.The number of male applicants is much higher than female applicants.
- 2.This indicates that more male applicants applied for loans compared to female applicants in this dataset.



Education Distribution

Insights:

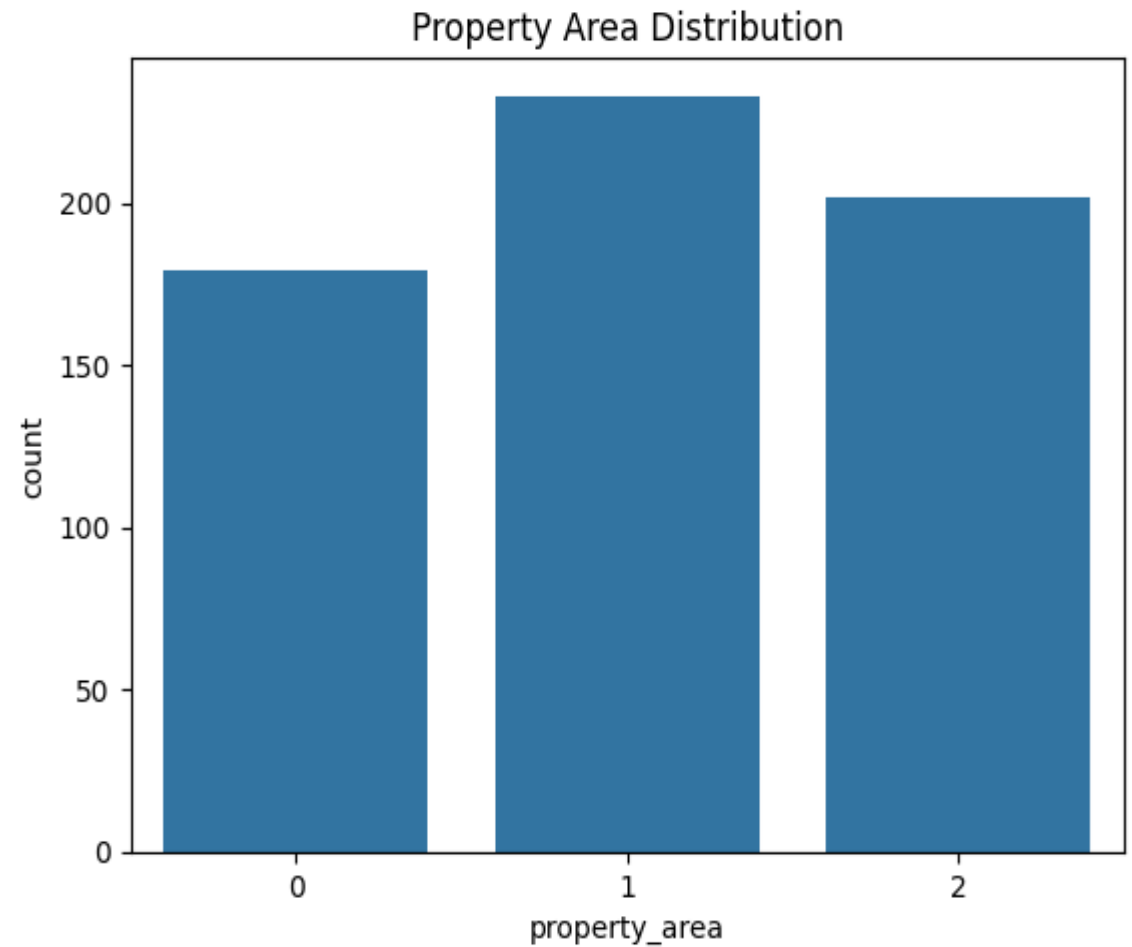
1. Graduate applicants are more than non-graduate applicants.
2. This suggests that educated applicants are more likely to apply for loans.
3. Education may also influence loan approval decisions.



Property Area Distribution

Insights:

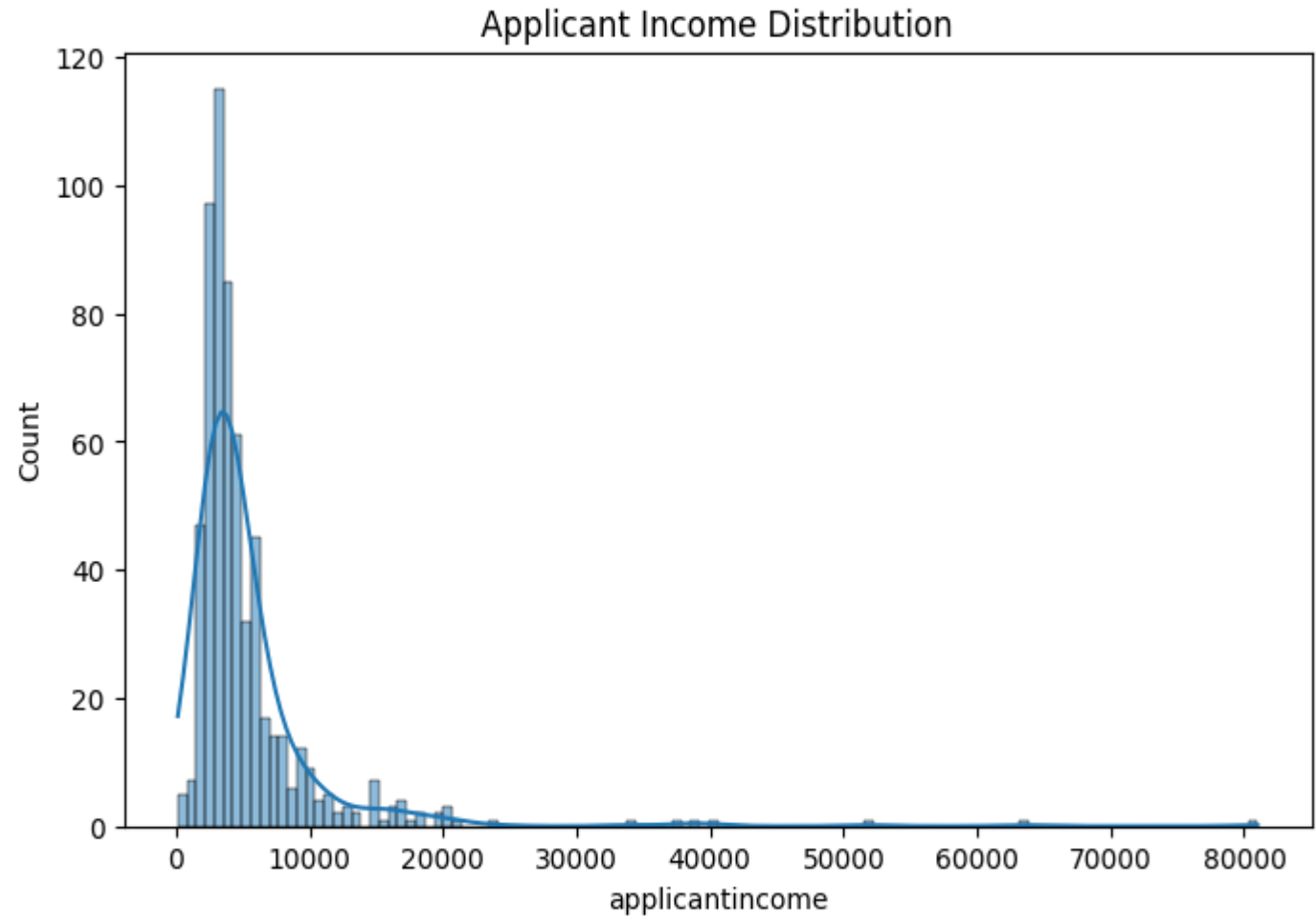
1. Most applicants belong to Semi urban areas, followed by Urban and Rural areas.
2. This shows that semiurban applicants are the most common in the dataset.
3. Location may affect loan approval patterns



Applicant Income Distribution

Insights:

1. Most applicants have income in the lower to middle income range.
2. Only a few applicants have very high income, creating a right-skewed distribution.
3. This means income is not evenly distributed.



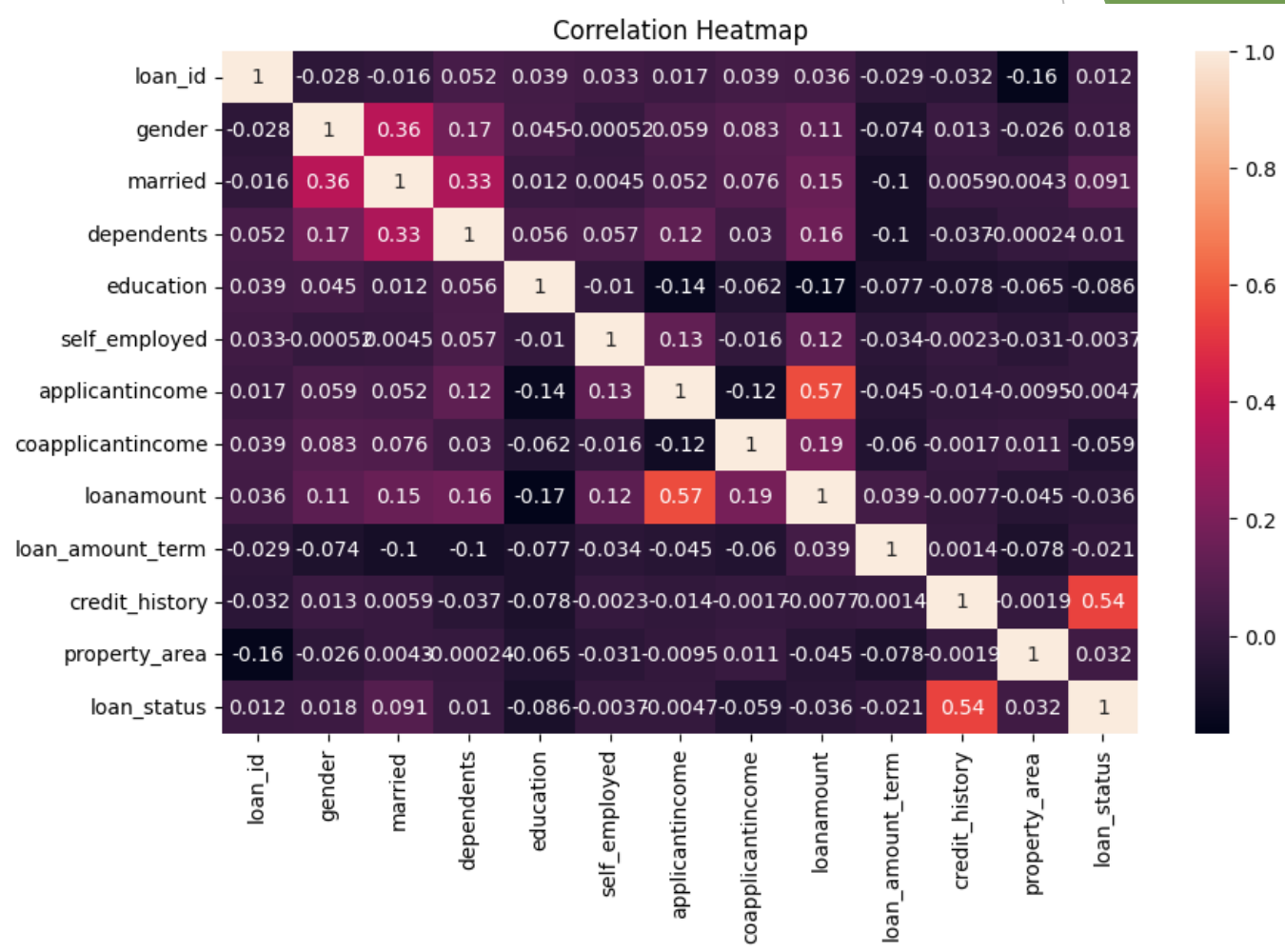
Correlation Heatmap

Insights:

1. Credit history shows a strong relationship with loan approval.

2. Applicant income and loan amount also show some relationship.

3. This indicates that credit history is one of the most important factors for loan approval prediction.



Data Splitting

The dataset was divided into:

- ↴ Training Data (80%)
- ↴ Testing Data (20%)
- ▶ Evaluates Training Model for better performance
- ▶ Trains the data for model training
- ▶ The split was performed using Scikit-learn's `train_test_split()` function.

Machine Learning Algorithm Used

- ▶ 1. Logistic Regression

- ▶ A statistical classification algorithm used for predicting binary outcomes.
- ▶ In this project, it predicts whether a loan will be approved or rejected.
- ▶ It is simple, fast, and effective for classification problems.

- ▶ 2. Decision Tree Classifier

- ▶ A tree-based algorithm that makes decisions using a series of conditions.
- ▶ It splits the dataset into branches based on important features.
- ▶ Easy to understand and visualize.

- ▶ 3. Random Forest Classifier

- ▶ An ensemble learning algorithm that combines multiple decision trees.
- ▶ Makes predictions through majority voting.
- ▶ Reduces overfitting and improves accuracy.
- ▶ Achieved the best performance in this project and was selected as the final model.

Model Comparison And Selection

Machine Learning Models Evaluated

Three machine learning algorithms were trained and tested on the Loan Approval Prediction dataset.

Model	Accuracy
Logistic Regression	78.86%
Random Forest Classifier	78.05%
Decision Tree Classifier	69.92%

Model Comparison



Logistic Regression

Accuracy: 78.86%

Highest accuracy among all models.

Simple and computationally efficient.

Convergence warning observed, indicating further optimization may improve performance.



Decision Tree

Accuracy: 69.92%

Easy to interpret and visualize.

Lower accuracy compared to other models.

More prone to overfitting.



Random Forest

Accuracy: 78.05%

Ensemble method combining multiple decision trees.

More robust and stable predictions.

Better generalization on unseen data.

Model Evaluation

- ▶ Model used: Random Forest Classifier

Model Performance

- ▶ Algorithm: Random Forest Classifier
- ▶ Test Accuracy: 78%
- ▶ Evaluation Metrics: Confusion Matrix, Precision, Recall, and F1-Score

Confusion Matrix:

[[18 25]

[2 78]]

<u>Actual/Predicted Result</u>	<u>Count</u>
Rejected Loan Correctly Rejected	18
Rejected Loan Wrongly Approved	25
Approved Loan Wrongly Rejected	2
Approved Loan Correctly Approved	78

Classification Report

Important Values:

For Approved Loans (Class 1)

- ↓ Precision = 0.76
- ↓ Recall = 0.97
- ↓ F1-score = 0.85

Observation:

The recall value is very high, which means the model successfully identifies most approved loan applications.

For Rejected Loans (Class 0)

- ↓ Precision = 0.90
- ↓ Recall = 0.42
- ↓ F1-score = 0.57

Observation:

The recall is lower, meaning the model misses some rejected loan applications.

Hyper Parametric Tunning

► Objective

To optimize the Random Forest model by finding the best parameter values and improving model performance.

Technique Used

.GridSearchCV with 5-Fold Cross Validation

Parametric Testing

Hyper Parametre	Value Tested
n_estimator	50,100,200
Max_depth	5,10,none

Best value of n_estimator is 100 and max_depth is 5

The improved Accuracy after hyper parametric tuning is 81%.
It has increased accuracy of 5% than the previos accurate value

Result

.Among all models, Random Forest gave better prediction accuracy because it handles multiple features efficiently and reduces overfitting.

.Credit History was found to be one of the most important features affecting loan approval.

. The final accuracy after Hyper parametric Test is 81%.

.Thus the project successfully predicts whether a loan should be approved or rejected.

App Deployment

.We have created an app for Loan Approval Prediction.

.We created the app using Streamlit.

.There were two files for the app where one file had the code for the app and another file had requirements.

.Libraries used for the app

1.Streamlit

2.Pandas

3.Pickle

.Libraries used for the requirement

1.Pandas

2.Streamlit

3.Numpy

4.Scikit-learn

App Prediction

 **Loan Approval Prediction System**

Enter applicant details below

Gender

Male

Married

Yes

Dependents

0

Education

Graduate

Self Employed

Yes

Applicant Income

20000

-

+

Coapplicant Income

5000

-

+

Loan Amount

95000

-

+

Loan Term

56

-

+

Credit History

1

Property Area

Semiurban

Predict

☒ Loan Approved

 **Loan Approval Prediction System**

Enter applicant details below

Gender

Male

Married

No

Dependents

2

Education

Graduate

Self Employed

No

Applicant Income

10000

-

+

Coapplicant Income

25000

-

+

Loan Amount

95000

-

+

Loan Term

56

-

+

Credit History

0

Property Area

Semiurban

Predict

☐ Loan Rejected

Future Scope

- ▶ Future Enhancements
 - ▶ ♦ Use Larger Datasets
 - ▶ Train the model on larger and more diverse loan application datasets.
 - ▶ Improve prediction accuracy and reliability.
 - ▶ ♦ Advanced Machine Learning Models
 - ▶ Implement advanced algorithms such as XGBoost, Gradient Boosting, and Neural Networks.
 - ▶ Compare their performance with Random Forest.
 - ▶ ♦ Real-Time Banking Integration
 - ▶ Integrate the system with banking databases and loan management systems.
 - ▶ Enable real-time loan approval decisions.
 - ▶ ♦ Enhanced Feature Selection
 - ▶ Include additional applicant information such as employment history, credit score, transaction records, and repayment behavior.

Conclusion

Successfully developed a Loan Approval Prediction System using Machine Learning techniques.

Performed data preprocessing, feature encoding, and model training on the loan dataset.

Evaluated multiple machine learning algorithms and selected the most suitable model.

Deployed the final model using Streamlit for real-time prediction and user interaction.

- ▶ Project Achievements
- ▶ ☒ Successfully analyzed and preprocessed 614 loan application records
- ▶ ☒ Implemented and compared Logistic Regression, Decision Tree, and Random Forest models
- ▶ ☒ Achieved approximately 78% prediction accuracy
- ▶ ☒ Identified Random Forest as the most reliable model for loan approval prediction
- ▶ ☒ Developed an interactive Streamlit web application
- ▶ ☒ Demonstrated a practical use of Machine Learning in the banking and finance sector

THANK YOU